Assignment 2 (Session 2)

1 -- List cleanup: Given [4, 2, 4, 1, 3], sort it and print its length.

2 — List operations: Start with [10, 20, 30]. Replace the middle value, append 40, and extend the list with [50, 60].

3 — Slicing: Given names = ["Aman", "Riya", "Kabir", "Sara"], print the first item, the last two items, and the first three items.

4 — Nested lists: Given marks = [[85, 91, 89], [78, 81, 86], [62, 75, 77]], print the second student's marks and the last student's second mark.

5 — Tuple access: Given colours = ("red", "green", "blue"), print the last item and a slice. Then try changing the first item and observe what happens.

6 — Sequence comparison: Create a list, tuple, and string. For each one, print its length, its item at index 1, and a slice from index 1 onward.

7 — Dictionary update: Create a dictionary for a student. Add a branch key, change an existing value, print its keys and values, then delete one key.

8 — Set algebra: Find the common values of {1, 2, 3, 4} and {3, 4, 5, 6}. Then print their union and difference.

9 — String methods: For text = " Python Bootcamp! ", remove the surrounding spaces, convert the result to lowercase, and check whether "Python" appears in the original string.

10 — String formatting: Ask the user for their name and print Welcome, <name>! using an f-string.

11 — for loop: Print each item in a list and then print that item multiplied by 2.

12 — range(): Use range() to print the even numbers from 2 through 20.

13 — while loop: Print 5, 4, 3, 2, 1, then print Go! exactly once.

14 — Loop + collection: Use a loop to calculate the total of [10, 20, 30, 40, 50].

15 — Loop choice: Print the items of a list using both a for loop and a while loop.